

Adilet Shynggys

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RESEARCH INTERESTS

My research examines how microscale interfacial phenomena — capillary and contact-line dynamics, wettability, and two-phase transport — govern macroscale power-system performance, from heat-exchanger efficiency to reactor thermal margins against subcooled boiling, departure from nucleate boiling, and critical heat flux, within broader interests spanning heat transfer, thermodynamics, and nuclear energy systems.

RESEARCH EXPERIENCE

Capillary-Driven Surface Engineering for Power-Plant Efficiency

August 2026 – Present

Department of Mechanical and Aerospace Engineering, Nazarbayev University • Advisor: Prof. Amgad Salama

Improving combustion-chamber and heat-exchanger efficiency in power plants by engineering capillary-driven surface structures that passively transport liquid slugs and droplets along heat-exchanger pipes, reducing flow maldistribution and heat loss without external pumping.

- Designing tapered SU-8 microchannels, with various width gradients on the order μm s for photolithographic fabrication on silicon: the taper creates a Laplace-pressure asymmetry intended to passively drive wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- Planning experiments to measure droplet and liquid-slug velocity across a range of taper geometries, to characterize how wettability and non-wettability govern transport behavior.
- Connecting this geometry-driven transport to passive liquid return in two-phase thermal management and to rewetting in heated channels.

Micro-Energy-Harvesting Fabrication and Characterization

January – December 2025

Energy Conversion Laboratory, Nazarbayev University

Applying the physical and surface properties of low-cost, simple materials toward sustainable, micro-scale green energy generation via triboelectric charge transfer.

- Fabricated triboelectric nanogenerators for self-powered wearable sensing (e.g., robotic-hand and haptic feedback systems, hiking/outdoor gear) via electrospinning and silicone casting, raising output power density by 15–20%.
- Characterized performance across humidity and cyclic durability testing; benchmarked maximum power output per unit surface area, peak open-circuit voltage, and short-circuit current across candidate applications.
- Built the electrical characterization workflow (load sweeps, data acquisition) in OriginPro and MATLAB, contributing the resulting data and analysis to a co-authored article in *Small* and an award-winning conference poster.

EDUCATION

Nazarbayev University

B.Eng. in Mechanical and Aerospace Engineering

Astana, Kazakhstan

Expected June 2027

- GPA: 3.86 / 4.00 — ranked 1st in cohort; Dean's List, four consecutive semesters.
- Relevant coursework: Fluid Mechanics; Engineering Thermodynamics; Numerical Methods; Differential Equations and Linear Algebra; Engineering Dynamics I–II; Structural Mechanics; Materials and Manufacturing; CAD. In progress: Heat and Mass Transfer; Mechanical Systems Design; Aerospace Propulsion.

The University of Hong Kong

Visiting Exchange Student, Department of Mechanical Engineering

Hong Kong SAR

Spring 2026

- Coursework in Energy Conversion Systems, Design and Manufacture, and Additive Manufacturing; completed two design projects during the exchange (see Engineering Design Projects) and collaborated with Prof. Wei Yan on curriculum design for an undergraduate course.

PUBLICATIONS AND CONFERENCE PRESENTATIONS

- [1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al. “Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator,” *Small*, 2026, art. e74486. <https://doi.org/10.1002/smll.74486>
- [2] **A. Shynggys**, S. Yavari, G. Kalimuldina. “Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors.” Presented as a poster at the Belt and Road Initiative Conference, Astana, 2025 (**Best Research Poster Award**), and as an abstract at the 13th International Conference on Nanomaterials and Advanced Energy Storage Systems (INESS-2025), Nazarbayev University, Astana, Aug. 2025.

ENGINEERING DESIGN PROJECTS

Reduced-Order Design and Analysis of a VVER-1200 Reactor Core

August 2026 – Present

Senior Capstone Project, Nazarbayev University • Advisor: Prof. Amgad Salama • Team of 4 students, 1 supervisor

- Leading a four-student team designing a VVER-1200 core from scratch: converting the required electric power output to thermal power via cycle efficiency, then using a target fuel burnup (in GWd per tonne heavy metal) to size the number of fuel pellets, rods, and bundles needed, arriving at a hexagonal core configuration.
- Developing physics-based reduced-order models coupling neutronic and thermal-hydraulic sub-models — multiplication factor, power distribution, coolant temperature, pressure drop, heat transfer, and margins to subcooled boiling and departure from nucleate boiling — each feeding the other.
- Verifying and validating the model against published reference data and benchmark problems, enabling rapid parametric design studies and preliminary safety assessment.

Other Engineering Design Projects

2025 – Spring 2026

- *Ergonomic Train-Driver Seat Design* (Mechatronics Engineering Case Competition, sponsored by Wabtec Corporation; team of 3, 1st Place): designed and structurally analyzed a driver's seat for 14-hour locomotive shifts, certifying it against GOST 33330-2015 under static/dynamic fatigue loading.
- *Multi-Stage Transmission (Gearbox) Design* (University of Hong Kong, Prof. Wei Yan): sized shafts using the ASME combined bending-torsion criterion and selected gear modules, keyways, and bearings to meet the project's specified safety margin and fatigue life.
- *Interactive Public Art Installation* (University of Hong Kong, Prof. Homayoon Kazerooni, UC Berkeley): modeled flower-petal forms in SolidWorks to specified dimensional requirements, arranging multiple petals to avoid geometric collisions and verifying structural stability, then 3D-printed each piece on a Bambu Lab A1.

INDUSTRY EXPERIENCE

Product Support Engineering Intern

June – July 2025

Wabtec Corporation — Kazakhstan

- Analyzed thermal performance of locomotive traction motors and supported root-cause review of field temperature excursions.
- Audited shop-floor manufacturing against engineering drawings with a fellow intern, reconciling GOST (shop) with ASME (Wabtec's U.S. standard).
- Traced an insulation-usage inconsistency across units to climate differences between export markets (e.g., uninsulated units for India vs. insulated units for Kazakhstan's winters).

AWARDS AND HONORS

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| • Presidential Scholarship of the Republic of Kazakhstan — national merit scholarship | 2026–2027 |
| • NU President's Scholarship — university merit scholarship | 2024–2025 |
| • Yessenov Foundation Scholarship — top 20 of 500+ applicants | 2026–2027 |
| • Best Research Poster, Belt and Road Initiative Conference | 2025 |
| • 1st Place, Mechatronics Engineering Case Competition (sponsored by Wabtec) | 2025 |
| • Champion, NU AIChE Engineering Challenges | 2025 |

TECHNICAL SKILLS

Programming	MATLAB, Python, Mathematica, $\text{\LaTeX}$
Modeling & Numerics	Reduced-order modeling of nuclear reactor core physics; validation against published reference data
Simulation & CAD	SolidWorks (CSWA certified), OriginPro
Experimental	Electrospinning, silicone casting, 3D printing, electrical device characterization, photolithographic microfabrication
Design Codes	ASME shaft design, GOST 33330-2015
Languages	Kazakh (native), Russian (fluent), English (IELTS Overall Band 8.0)

TEACHING, LEADERSHIP, AND SERVICE

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| • Physics Tutor, Nazarbayev University — 20+ first-year students. | 2023–2024 |
| • Cohort Representative, Mechanical and Aerospace Engineering. | 2024–2026 |
| • Organizer, Charity Trail Run — 150 runners, 600,000 KZT (~\$1,300) raised. | 2025 |
| • Volunteer, HKU Muslim Society and Mereke Charity Foundation. | 2025–2026 |